

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Senior full stack engineer with eight years building end-to-end web platforms across fintech and enterprise environments. Owns features from database schema through polished UI, with deep experience in TypeScript, React, Node.js, and PostgreSQL on AWS. Integrates LLM APIs into production with prompt validation, cost controls, and graceful degradation. Influences architecture decisions, mentors engineers, and ships resilient systems using modern CI/CD and agent-assisted development workflows.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led **end-to-end delivery** of investor-facing portfolio dashboards using **React**, **TypeScript**, and **Node.js**, reducing report generation latency by 40% through query optimization and caching layers.
- Designed and implemented **REST and GraphQL APIs** serving 50K+ monthly requests, establishing versioning standards and **OAuth-based access control** across internal and client-facing services.
- Built **AI-assisted analytics features** integrating **LLM APIs** with structured prompt templates, output validation, and **cost-aware routing** to deliver portfolio summaries with graceful fallback when models exceed latency thresholds.
- Deployed and operated **cloud-native services on AWS** (ECS, RDS, S3, CloudFront), achieving 99.9% uptime through auto-scaling policies, health checks, and runbook-driven incident response.
- Optimized **PostgreSQL** schemas and indexes for multi-tenant fund data, cutting slow-query incidents by 60% and enabling faster onboarding of new investment strategies.
- Drove **CI/CD pipeline** improvements with automated testing, preview environments, and **agentic code review workflows** using Cursor, increasing deployment frequency while maintaining zero critical regressions in production.
- Partnered with product leadership to translate ambiguous requirements into **scalable architecture proposals**, influencing a 12-month platform roadmap for marketplace-style data sharing between fund managers and LPs.
- Mentored **three junior engineers** through pairing, structured code reviews, and career planning, raising team PR throughput and establishing shared patterns for **secure coding** aligned with **OWASP** practices.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **full stack trading and risk reporting tools** using JavaScript, internal frameworks, and **SQL databases**, supporting analysts across fixed income and equities desks in New York.
- Implemented **secure API integrations** between legacy banking systems and modern web interfaces, enforcing encryption at rest, role-based permissions, and audit logging for **regulatory compliance**.
- Optimized batch **data pipelines** processing millions of daily market records, reducing overnight job durations by 35% through parallelization and index tuning.
- Contributed to **architecture reviews** for a client portal modernization initiative, advocating pragmatic trade-offs between microservice decomposition and delivery speed.
- Built responsive **React** frontends for internal dashboards consumed by 200+ analysts, improving data discoverability and cutting time-to-insight for daily risk assessments.
- Automated deployment workflows with **CI/CD tooling**, standardizing build pipelines across three application teams and reducing manual release steps by 70%.
- Collaborated with **cross-functional stakeholders**—product owners, compliance, and infrastructure—to ship features under strict data privacy requirements including **GDPR-aligned** data handling.
- Led **code review** sessions and documented engineering standards, mentoring incoming summer analysts and establishing reusable component libraries for the technology group.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **React components** for a restaurant compliance tracking platform, improving mobile responsiveness and reducing page load times through lazy loading and asset optimization.
- Developed **Node.js microservices** handling receipt digitization workflows, integrating third-party OCR APIs with retry logic and structured error handling.
- Wrote **PostgreSQL migrations** and queries supporting multi-location restaurant data, enabling operators to filter compliance records across 100+ venues.
- Participated in **Agile ceremonies** and paired with senior engineers on REST API design, gaining hands-on experience with authentication middleware and input validation.
- Added **unit and integration tests** to critical payment reconciliation paths, increasing coverage on core billing modules and preventing regressions during weekly releases.
- Assisted with **AWS EC2** deployment scripts and environment configuration, documenting runbooks that shortened onboarding for subsequent intern cohorts.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed static and dynamic web pages for a nonprofit research foundation using **HTML**, **CSS**, and **JavaScript**, delivering an accessible site aligned with organizational branding guidelines.
- Integrated a **CMS-backed content workflow** allowing non-technical staff to publish research updates without developer intervention, reducing publication turnaround from days to hours.
- Implemented **responsive layouts** ensuring consistent rendering across desktop and mobile browsers for donor and event registration flows.
- Configured **Google Analytics** and basic SEO metadata, improving visibility of foundation publications and tracking engagement on key landing pages.
- Collaborated with foundation leadership to gather requirements and iterate on **UX improvements** for newsletter signup and contact forms with client-side validation.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Responsive Design, Component Architecture, GraphQL Client Integration

Backend: Node.js, Express, REST APIs, GraphQL, OAuth 2.0, Microservices, Golang (working proficiency), API Design

Databases: PostgreSQL, SQL, Schema Design, Query Optimization, Indexing, ClickHouse (analytical workloads), Redis

Cloud & DevOps: AWS (ECS, RDS, S3, CloudFront, EC2, Lambda), Docker, CI/CD, Terraform, GitHub Actions, Infrastructure as Code

Monitoring & Tools: Datadog, CloudWatch, Sentry, Git, Cursor, Claude Code, Postman, Jest, Cypress, ESLint, Prettier

Workflow & PM: Agile/Scrum, Code Review, Technical Mentorship, Architecture Documentation, Jira, Confluence, Stakeholder Communication